

Cu-14.3Al-4.3Ni wt% - composition-only audit

Composition to crystal structure, phase stability, properties, and additive-manufacturing screening

Report CAN-PRIMARY-COMPOSITION-ONLY | 2026-09-09 00:00 UTC

Scientific interpretation rule. Physics calculations, ML predictions, engineering screens, and experimental measurements are distinct evidence classes. Unavailable results are not replaced with assumed values. Review each section's provenance, confidence, and limitations before making experimental or manufacturing decisions.

Composition and descriptors

Physics calculated | Confidence: Deterministic composition conversion; descriptor uncertainty follows input uncertainty | Provenance: composition.converter.AlloyComposition and composition.descriptors.compute_descriptors

Result	Value
Input Wt Percent	Cu: 81.4; Al: 14.3; Ni: 4.3
Weight Percent	Cu: 81.4; Al: 14.3; Ni: 4.3
Atomic Percent	Cu: 67.9838; Al: 28.128; Ni: 3.8882
Mole Fraction	Cu: 0.679838; Al: 0.28128; Ni: 0.038882
Hume Rothery E A	Unavailable [not available]
Valence Electron Concentration	8.71088 electrons/atom [physics calculated]
Density Estimate	6.72754 g/cm^3 [engineering screening]

Limitations: Composition conversion assumes the atomic-weight constants documented by the composition module. | Density and electronic descriptors, if present, are mixture-rule descriptors rather than measured bulk properties.

Thermodynamics and phase stability

Unavailable | Confidence: Not applicable | Provenance: No thermodynamic database supplied

Result	Value
Reason	Upload a licensed/open Cu-Al-Ni TDB containing compatible Cu, Al, and Ni phases to calculate equilibrium.

Limitations: Upload a licensed/open Cu-Al-Ni TDB containing compatible Cu, Al, and Ni phases to calculate equilibrium.

Phase assessment

Unavailable | Confidence: Not applicable | Provenance: No thermodynamic database supplied

Result	Value
Phase Reference Catalog	5 structured records: beta, beta1, 18R, 2H, gamma2 (see JSON export for full metadata)
Risks	gamma2: Requires a database-calculated gamma2 phase fraction or a cited screening profile.; brittleness: Requires measured properties or a cited engineering screening profile.; segregation: Requires Scheil/kinetic results or spatial experimental composition data.

Limitations: Upload a licensed/open Cu-Al-Ni TDB containing compatible Cu, Al, and Ni phases to calculate equilibrium.

Crystallography

Unavailable | Confidence: Not applicable | Provenance: crystallography.crystal_database literature registry

Result	Value
Reason	No complete CIF was supplied. Literature metadata are shown, but atomic coordinates are required for 3D visualization and structure-factor XRD.
Literature References	3 structured records: 18R, 2H, GAMMA2 (see JSON export for full metadata)
Load Errors	

Limitations: No complete CIF was supplied. Literature metadata are shown, but atomic coordinates are required for 3D visualization and structure-factor XRD.

Diffraction / XRD

Unavailable | Confidence: Not applicable | Provenance: Not executed

Result	Value
Reason	Diffraction simulation requires a complete validated CIF.

Limitations: Diffraction simulation requires a complete validated CIF.

Machine-learning property predictions

Unavailable | Confidence: Not applicable | Provenance: No authorized training dataset supplied

Result	Value
Reason	Upload a provenance metadata JSON and compatible experimental CSV to train transformation/property models.

Limitations: Upload a provenance metadata JSON and compatible experimental CSV to train transformation/property models.

Experimental characterization

Unavailable | Confidence: Not applicable | Provenance: No experimental data supplied

Result	Value
Reason	Upload XRD, EDS, DSC, or SEM data to run experimental characterization.

Limitations: Upload XRD, EDS, DSC, or SEM data to run experimental characterization.

Additive-manufacturing screening

Unavailable | Confidence: Not applicable | Provenance: No engineering-screening normalization profile supplied

Result	Value
Reason	A documented AM normalization profile is required before a comparable 0-10 suitability score can be calculated.

Limitations: A documented AM normalization profile is required before a comparable 0-10 suitability score can be calculated.

This decision-support report does not replace thermodynamic database licensing, instrument calibration, metallographic validation, qualified process development, or expert review.